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SARDAR PATEL UNIVERSITY
BCA Semester-4 Examination, 2022
US04CBCA27 || Operating Systems
7th April, 2022.

Time : 03.00 p.m. – 05.00 p.m.

Marks: 70

Q-1 Select the appropriate option for the following questions:

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- 1) _____ is NOT a valid Process state.
A. Ready C. Running
B. File D. Terminate
- 2) Time-sharing system also known as _____.
A. Real-time C. Multi-user
B. Multi-tasking D. None
- 3) _____ page replacement algorithm suffers from Belady's anomaly.
A. FIFO C. LRU
B. Optimal D. None
- 4) Swapping of process requires _____.
A. Operating System C. Backing Store
B. Main Memory D. All of these
- 5) Each process has a segment of code called _____.
A. important Section C. Critical Section
B. Mutual Section D. None
- 6) _____ is not a valid LINUX file system.
A. EXT2 C. EXE
B. EXT D. All of them
- 7) _____ command is use to display list of all users currently login on a server.
A. grep C. search
B. find D. None
- 8) _____ command is use to move file from one location to another.
A. cp C. rm
B. mv D. None
- 9) _____ scheduling algorithm gives minimal average waiting time.
A. FCFS C. Round-robin
B. SJF D. Priority
- 10) Demand Paging is _____.
A. Paging with swapping C. Swapping with fragmentation
B. Paging without Swapping D. Swapping without fragmentation

Q-2 Tick (TRUE or FALSE) and Fill in the blanks.

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- 1) A Program in execution is known as _____.
- 2) In demand paging, Process which is in main memory is denoted by _____ bit.
- 3) The situation when waiting process never change its state because of no availability of resources is called _____.
- 4) _____ command is use to change a permission of a file.

P.T.O.

- 5) PCB = Program Control Block . [TRUE/FALSE]
- 6) Optimal scheduling algorithm gives minimum Page Faults. [TRUE/FALSE]
- 7) First UNIX was developed by a Ken Thompson. [TRUE/FALSE]
- 8) In LINUX, Case statement ends with end case. [TRUE/FALSE]

Q-3 Give short answers of the following questions: (ANY TEN)

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- 1) Explain Time sharing system in brief.
- 2) Draw the diagram of PCB.
- 3) List all layers of Layered structure of OS.
- 4) Explain Virtual memory in short.
- 5) Calculate Page faults using FIFO algorithm for following reference string: (Number of Frames = 3)
Reference string = 7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1
- 6) What is Compaction? For what purpose it will use?
- 7) Explain any two necessary conditions of Critical section.
- 8) Why is LINUX an open source?
- 9) Explain algorithm-1 for two-process solution.
- 10) Explain if statement in LINUX.
- 11) Explain cut command in short.
- 12) Explain ls -l command.

Q-4 Give Long answers of the following questions: (ANY FOUR)

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- 1) List out various types of OS. Explain Real Time Operating System.
- 2) Describe FCFS scheduling with advantages and disadvantage.
- 3) Explain Swapping of two processes with suitable example.
- 4) Explain FIFO page replacement algorithm with advantages and disadvantages.
- 5) What do you mean by Deadlock? Explain all necessary conditions for occurrence of deadlock.
- 6) What is LINUX? Explain basic features of LINUX Operating System.
- 7) Explain for construct in Linux.
- 8) What is FAP? Explain various ways to change permission on a file.

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(2)